

Smart Cities: AI-Based Vehicle Tracking and License Plate Recognition System

PROBLEM STATEMENT:

Modern smart cities face increasing traffic congestion, law violations, and inefficient monitoring systems.

Manual surveillance is costly, slow, and error-prone.

This project aims to build an AI-powered automated traffic monitoring system that can detect vehicles, count them, and recognize license plates in real-time for better traffic management and law enforcement.

OBJECTIVES:

- Detect vehicles (car, bike, truck, bus) in real-time video streams
- Count vehicles crossing a virtual line
- Extract license plate numbers using OCR, if vehicle cross the virtual line
- Store vehicle type, license plate and timestamp in a database
- Provide analytics for smart city traffic management

SYSTEM ARCHITECTURE:

Video Input



Frame Extraction (OpenCV)



YOLOv8 Vehicle Detection Model (pretrained on COCO dataset)



DeepSORT / ByteTrack Tracking



Line Crossing Detection Module



License Plate Detection Model (YOLO ANPR)



OCR Engine (EasyOCR / Tesseract)



Database Storage (SQLite / PostgreSQL)



Streamlit Dashboard / FastAPI Service

STEP-BY-STEP WORKFLOW:

1. Data Input

- Load CCTV footage or traffic dataset videos
- Extract frames using OpenCV

2. Vehicle Detection

- Use pretrained YOLOv8 model (COCO dataset)
- Detect:
 - Car
 - Bike
 - Truck
 - Bus

3. Tracking

- Apply DeepSORT or ByteTrack
- Assign unique ID to each vehicle
- Ensure same vehicle is not counted multiple times

4. Line Crossing

- Define a virtual line on road
- Detect centroid crossing event
- Increment vehicle counter per class

5. License Plate Detection

- Apply YOLO-based license plate detection model
- Trigger only when vehicle crosses line

6. OCR

- Crop license plate region
- Apply EasyOCR / Tesseract
- Convert image → text (plate number)

7. Database (SQLite)

Store structured records:

- Vehicle type
- License plate number
- Timestamp of event

8. Dashboard

- Live vehicle count (class-wise)
- License plate logs
- Traffic analytics graphs
- Historical reports

DATASET LINK (License plate dataset):

<https://universe.roboflow.com/roboflow-universe-projects/license-plate-recognition-rxg4e>

EXPECTED OUTPUT:

- Vehicle detection and counting
- License plate recognition
- Traffic analytics dashboard
- Stored logs in database

Example Database Output

🔗 “10 trucks crossed the line”

Database will store:

Vehicle Type	Plate Number	Timestamp
Truck	ABC-123	10:01 AM
Truck	XYZ-456	10:02 AM
Truck	DEF-789	10:05 AM

Sample Videos for Testing (Links)

<https://drive.google.com/drive/folders/11n5u1B3BAppISJl7OgLNII3shY9FnLsV?usp=sharing>

Deliverables:

```
smart-traffic-anpr/
├── data/
│   ├── videos/
│   │   └── traffic.mp4
│   └── images/
├── notebooks/
│   ├── vehicle_detection.ipynb
│   ├── tracking_and_counting.ipynb
│   └── license_plate_ocr.ipynb
├── models/
│   ├── vehicle_model.pt
│   └── plate_model.pt
├── database/
│   ├── traffic.db
│   └── schema.sql
├── app/
│   └── streamlit_app.py
├── output/
│   ├── results_video.mp4
│   └── logs.csv
├── requirements.txt
└── README.md
```

TECH STACK:

Python, OpenCV, YOLOv8, DeepSORT, EasyOCR, FastAPI, Streamlit

IMPACT:

This system helps smart cities automate traffic monitoring, reduce manual surveillance, and improve law enforcement efficiency.